Multi-Asset Portfolio Attribution System

1. Executive Summary

This project establishes an automated pipeline to analyze the performance of a multi-asset investment portfolio (WTI Oil, Gold, and NVDA). By integrating **Python-driven quantitative strategies** with a **PostgreSQL-based Brinson-Fachler attribution model**, the system successfully identifies the drivers of a **0.15 Total Alpha** achieved during the Q1 2024 backtest period.

2. Methodology & System Architecture

A. Data Engineering (ETL Pipeline)

- **Data Sourcing**: Real-market data is extracted via the yfinance API in Python, covering historical prices for Commodities and Equities.
- **Database Schema**: A relational star schema was implemented in **PostgreSQL**, featuring dimension tables for securities and benchmark metadata, and fact tables for daily market prices and portfolio holdings.

B. Quantitative Strategy: Risk Parity

Moving beyond static allocations, the system employs an **Inverse Volatility Weighting** (Risk Parity) strategy.

- **Logic**: Daily weights are dynamically adjusted based on the 20-day rolling standard deviation of returns.
- **Implementation**: Python computes these risk-adjusted weights to ensure that high-volatility assets (e.g., WTI Oil) do not disproportionately dominate the portfolio's risk profile.

C. Attribution Model: Brinson-Fachler

The core analytical engine utilizes SQL Common Table Expressions (CTEs) to decompose excess returns into two primary effects:

1. **Allocation Effect (A_i):** Impact of over/underweighting sectors relative to the benchmark.

$$A_i = (W_{p,i} - W_{b,i}) \times (R_{b,i} - R_{b,total})$$

2. **Selection Effect (S_i):** Impact of choosing specific assets that outperform their respective sector benchmarks.

$$S_i = W_{p,i} \times (R_{p,i} - R_{b,i})$$

3. Key Findings & Performance Analysis

The final dashboard reveals a robust performance profile:

- **Total Alpha**: 0.15 (15% excess return over the period).
- **Primary Driver**: The **Selection Effect** was the dominant contributor, particularly within the **Materials (Gold)** and **Technology (NVDA)** sectors.
- **Risk Management**: The dynamic rebalancing successfully smoothed the contribution from Energy, preventing extreme drawdowns during periods of high oil price volatility.

4. Technical Stack

- **Programming**: Python (Pandas, NumPy, SQLAlchemy).
- **Database**: PostgreSQL (Advanced SQL, Window Functions, CTEs).
- **Analytics**: Power BI (Data Modeling, Interactive Visualization).

5. Conclusion

This system bridges the gap between raw market data and actionable investment insights. It proves that a disciplined, risk-aware approach to asset allocation, combined with strong security selection, can generate significant Alpha in volatile market conditions.